<2018학년도 수학1 중간고사 예시 답안>

1.(1)

$f(x) = \sin(2\tan^{-1}x)$

$\tan^{-1}x = \theta$ 로 치환하자. $\tan\theta = x \quad (-\frac{\pi}{2} < \theta < \frac{\pi}{2})$

$\sin\theta = \frac{x}{\sqrt{1+x^2}}$, $\cos\theta = \frac{1}{\sqrt{1+x^2}}$

$\therefore f(x) = \sin(2\tan^{-1}x) = \sin 2\theta = 2\sin\theta\cos\theta = \frac{2x}{1+x^2}$

$\frac{2x}{1+x^2} = \frac{1}{3} \quad \Leftrightarrow x^2 - 6x + 1 = 0 \quad \Leftrightarrow x = 3 \pm 2\sqrt{2}$

(2)

$f(\sqrt{3}) = \frac{\sqrt{3}}{2} \qquad f'(\sqrt{3}) = -\frac{1}{4}$

접선의 방정식 : $y - f(\sqrt{3}) = f'(\sqrt{3})(x - \sqrt{3})$

$$\Rightarrow y = -\frac{1}{4}(x - \sqrt{3}) + \frac{\sqrt{3}}{2}$$

(3)

$f'(c) = \frac{2-2c^2}{(1+c^2)^2} = \frac{f(1)-f(0)}{1-0} = 1$

$c^4 + 2c^2 + 1 = 2 - 2c^2$

$c^4 + 4c^2 - 1 = 0$

$\therefore c^2 = -2 + \sqrt{5} \qquad c = \sqrt{-2+\sqrt{5}} \ (\because c \in (0,1))$

2.(1)

$A = \int 2\pi x ds$

$= \int_1^2 2\pi x \sqrt{1+(2x)^2}\, dx \qquad [1+4x^2 = t,\ 8xdx = dt]$

$= \int_5^{17} 2\pi \sqrt{t}\, \frac{1}{8} dt$

$= \frac{\pi}{6}(17\sqrt{17} - 5\sqrt{5})$

또는

$A = \int 2\pi x ds$

$= \int_1^4 2\pi \sqrt{y} \sqrt{1+(\frac{1}{2\sqrt{y}})^2}\, dy$

$= \int_1^4 \pi \sqrt{4y+1}\, dy$

$= \frac{\pi}{6}(17\sqrt{17} - 5\sqrt{5})$

(2)

$\cosh^{-1}x = y \qquad x = \cosh y$

$1 = \sinh y \frac{dy}{dx} \qquad \frac{dy}{dx} = \frac{1}{\sinh y}$

$y > 0$ 이므로 $\sinh y > 0$ 그러므로 $\sinh y = \sqrt{(\cosh x)^2 - 1}$

$\frac{dy}{dx} = \frac{1}{\sqrt{x^2-1}}$

3.(1)

$f'(x) = (3x^2+1)\cosh(x^3+x-2) > 0$ 이므로

증가함수이므로 일대일 대응하여 역함수가 존재한다.

(2)

$f(1) = 3 \qquad (f^{-1})'(3) = \frac{1}{f'(1)} = \frac{1}{4}$

$y = \frac{1}{4}(x-3)+1$

$\therefore y = \frac{1}{4}x + \frac{1}{4}$

(3)

$(f^{-1}f)'(x) = 1$

$(f^{-1})'(f(x))f'(x) = 1$

$(f^{-1})''(f(x))(f'(x))^2 + (f^{-1})'(f(x))f''(x) = 0$

$(f^{-1})''(3)(f'(1))^2 + (f^{-1})'(3)f''(1) = 0$

$(f^{-1})''(3) = \dfrac{-(f^{-1})'(3)f''(1)}{(f'(1))^2}$

$f'(x) = (3x^2+1)\cosh(x^3+x-2), f''(x) = 6x\cosh(x^3+x-2) + (3x^2+1)^2\sinh(x^3+x-2)$

이므로 $(f^{-1})''(3) = -\dfrac{3}{32}$ 이다.

4.(1)

$\lim\limits_{x\to\infty}(e^{3x}+x)^{\frac{3}{x}} = \lim\limits_{x\to\infty} y,\ \ln y = \dfrac{3}{x} ln(e^{3x}+x)$

$\lim\limits_{x\to\infty}\dfrac{3}{x}ln(e^{3x}+x) = \lim\limits_{x\to\infty}\dfrac{9(e^{3x}+1)}{(e^{3x}+x)} = \lim\limits_{x\to\infty}\dfrac{27(e^{3x})}{3(e^{3x}+1)} = \lim\limits_{x\to\infty}\dfrac{9}{(1+1/e^{3x})} = 9$

$(\because \dfrac{\infty}{\infty}$, 로피탈의 정리)

$\therefore \lim\limits_{x\to\infty} y = e^9$

(2)

$\lim\limits_{x\to 0}(1+x^2)^{\cot x} = \lim\limits_{x\to 0} y, \quad \ln y = \cot x \ln(1+x^2)$

$\lim\limits_{x\to 0}\cot x\ln(1+x^2) = \lim\limits_{x\to 0}\dfrac{\ln(1+x^2)}{\tan x} = \lim\limits_{x\to 0}\dfrac{2x/1+x^2}{\sec^2 x} = 0$

$(\because \dfrac{0}{0}$, 로피탈의 정리)

$\therefore \lim\limits_{x\to 0} y = e^0 = 1$

(3)

$\lim\limits_{x\to\infty}(x^2-\ln x) = \lim\limits_{x\to\infty}(\ln e^{x^2}-\ln x) = \lim\limits_{x\to\infty}(\ln(\dfrac{e^{x^2}}{x})) = \ln(\lim\limits_{x\to\infty}(\dfrac{e^{x^2}}{x})) = \infty$

$\lim\limits_{x\to\infty}(\dfrac{e^{x^2}}{x}) = \lim\limits_{x\to\infty}\dfrac{2xe^{x^2}}{1} = \infty \quad (\because \dfrac{\infty}{\infty}$, 로피탈의 정리)

5.(1)

$$\int \tan^{-1}x dx$$

$$= \int 1 \cdot \tan^{-1}x dx \qquad (f = \tan^{-1}x,\ f' = \frac{1}{1+x^2},\ g' = 1,\ g = x)$$

$$= x\tan^{-1}x - \int \frac{x}{1+x^2}dx$$

$$= x\tan^{-1}x - \frac{1}{2}ln\left|1+x^2\right| + C$$

$$= x\tan^{-1}x - \frac{1}{2}ln\,(1+x^2) + C$$

(2)

$$\int \tan^4 x\, dx$$

$$= \int \tan^2 x(\sec^2 x - 1)\,dx$$

$$= \int \tan^2 x \sec^2 x\, dx - \int \tan^2 x\, dx$$

$$(\because \int \tan^2 x \sec^2 x\, dx = \int u^2 du\ (\tan x = u) = \frac{1}{3}u^3 + C = \frac{1}{3}tan^3 x + C$$

$$\int \tan^2 x\, dx = \int (\sec^2 x - 1)\,dx = \tan x - x + C\ )$$

$$= \frac{1}{3}\tan^3 x - (\tan x - x) + C$$

$$\therefore \int_0^{\frac{\pi}{4}} \tan^4 x\, dx = \frac{1}{3} - 1 + \frac{\pi}{4} = \frac{\pi}{4} - \frac{2}{3}$$

(3)

$$\tan\frac{x}{2} = t \text{ 치환} \Rightarrow \cos x = \frac{1-t^2}{1+t^2},\ dx = \frac{2}{1+t^2}dt$$

$$\Rightarrow \int_0^1 \frac{2(1-t^2)}{(t^2+3)(t^2+1)}dt = \int_0^1 \frac{2}{(t^2+1)} + \frac{-4}{(t^2+3)}dt$$

$$= 2\left[\tan^{-1}(1) - \tan^{-1}(0)\right] - \frac{4}{\sqrt{3}}\left[\tan^{-1}(\frac{1}{\sqrt{3}}) - \tan^{-1}(\frac{0}{\sqrt{3}})\right]$$

$$= \frac{\pi}{2} - \frac{2\sqrt{3}}{9}\pi$$

(4)

$$\frac{2x^3+5x^2+8x+4}{(x^2+2x+2)^2} = \frac{Ax+B}{(x^2+2x+2)} + \frac{Cx+D}{(x^2+2x+2)^2}$$

$A=2, B=1, C=2, D=2$

$$\therefore \int \frac{2x^3+5x^2+8x+4}{(x^2+2x+2)^2}\,dx = \int \frac{2x+1}{(x^2+2x+2)}\,dx + \int \frac{2x+2}{(x^2+2x+2)^2}\,dx$$

$$= \int \frac{2x+2}{(x^2+2x+2)}\,dx - \int \frac{1}{(x^2+2x+2)}\,dx + \int \frac{2x+2}{(x^2+2x+2)^2}\,dx$$

$$= \ln(x^2+2x+2) - \tan^{-1}(x+1) - \frac{1}{x^2+2x+2} + C$$

6.

$$\int_0^{\frac{3}{5}} \frac{x^2}{\sqrt{9-25x^2}}\,dx = \lim_{t\to(\frac{3}{5})^-} \int_0^t \frac{x^2}{\sqrt{9-25x^2}}\,dx$$

$$\int_0^t \frac{x^2}{\sqrt{9-25x^2}}\,dx = \int_0^{\sin^{-1}(\frac{5t}{3})} \frac{(\frac{3}{5})^2\sin^2\theta}{3\cos\theta}\,\frac{3}{5}\cos\theta\,d\theta$$

$\left(x = \frac{3}{5}\sin\theta,\ dx = \frac{3}{5}\cos\theta\,d\theta,\ -\frac{\pi}{2} < \theta < \frac{\pi}{2} \right)$

$$= \int_0^{\sin^{-1}(\frac{5t}{3})} \frac{9}{125} \cdot \frac{1-\cos 2\theta}{2}\,d\theta = \frac{9}{250}\left(\sin^{-1}\frac{5t}{3} - \sin\alpha\cos\alpha\right) \quad \left(\because \sin^{-1}\frac{5t}{3} = \alpha\right)$$

$$= \frac{9}{250}\,\sin^{-1}\frac{5t}{3} - \frac{5t}{3}\sqrt{1-(\frac{5t}{3})^2}$$

$$\therefore \lim_{t\to(\frac{3}{5})^-} \frac{9}{250}\,\sin^{-1}\frac{5t}{3} - \frac{5t}{3}\sqrt{1-(\frac{5t}{3})^2} = \frac{9}{250}\sin^{-1}1 = \frac{9\pi}{500}$$

7.

$$\int_0^\infty \frac{1}{\sqrt{x}+x^2}\,dx = \int_0^1 \frac{1}{\sqrt{x}+x^2}\,dx + \int_1^\infty \frac{1}{\sqrt{x}+x^2}\,dx$$

$0 < x \le 1$, $0 \le \frac{1}{\sqrt{x}+x^2} \le \frac{1}{\sqrt{x}}$ 이고,

$\int_0^1 \frac{1}{\sqrt{x}}\,dx = \lim_{t\to 0^+}\int_t^1 x^{-\frac{1}{2}}\,dx = \lim_{t\to 0^+}(2-2\sqrt{t}) = 2$: 수렴

비교정리에 의해 $\int_0^1 \frac{1}{\sqrt{x}+x^2}\,dx$: 수렴한다.

$x \ge 1$, $0 \le \frac{1}{\sqrt{x}+x^2} \le \frac{1}{x^2}$ 이고,

$\int_1^\infty \frac{1}{x^2}dx = \lim_{t\to\infty}\int_1^t \frac{1}{x^2}dx = \lim_{t\to\infty}(-\frac{1}{t}+1) = 1$: 수렴

비교정리에 의해 $\int_1^\infty \frac{1}{\sqrt{x}+x^2}\,dx$: 수렴한다.

$\therefore \int_0^\infty \frac{1}{\sqrt{x}+x^2}\,dx$: 수렴

8.(1)

바깥쪽 반지름은 $r_0 = 1+\sqrt{x}$, 안쪽 반지름은 $r_i = 1+x^2$ 이므로

$$V = \pi\int_0^1 \{(1+\sqrt{x})^2 - (1+x^2)^2\}\,dx$$

$$= \pi\int_0^1 (2\sqrt{x} + x - 2x^2 - x^4)\,dx$$

$$= \pi(\frac{4}{3}+\frac{1}{2}-\frac{2}{3}-\frac{1}{5}) = \frac{29}{30}\pi$$

(2)

원통의 반지름은 $1+y$, 높이는 $\sqrt{y}-y^2$ 이므로

$$V = \int_0^1 2\pi(1+y)(\sqrt{y}-y^2)\,dy$$

$$= \int_0^1 2\pi(\sqrt{y} - y^2 + y\sqrt{y} - y^3)\,dx$$

$$= 2\pi(\frac{2}{3}-\frac{1}{3}+\frac{2}{5}-\frac{1}{4}) = \frac{29}{30}\pi$$